

DATA ANALYSIS PROJECT

# Biodiversity in National Parks

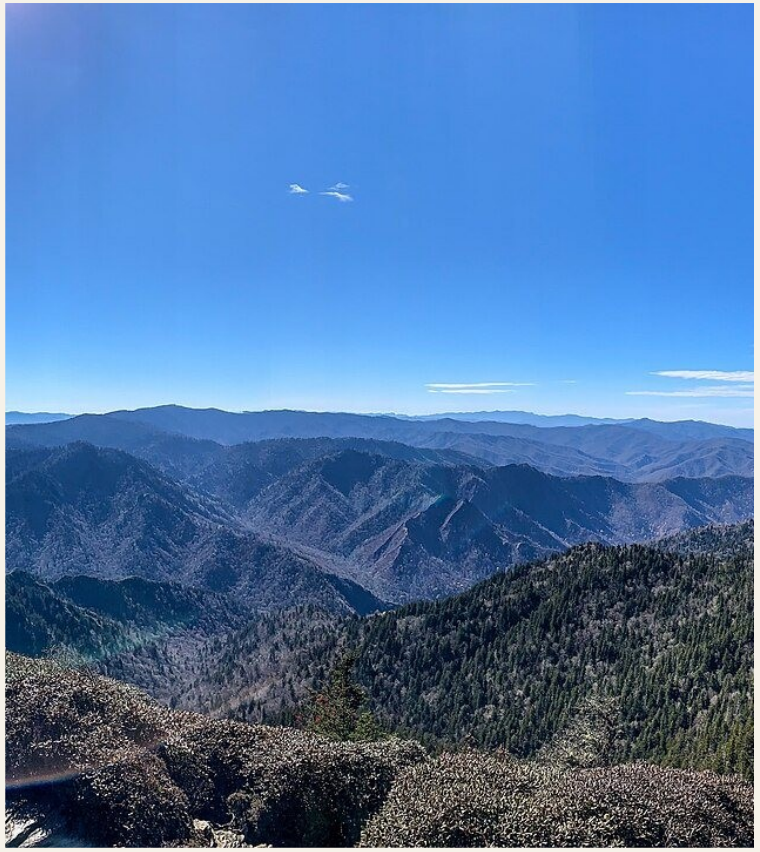
Conservation status of species across four national parks

**Rolin**

September 19

# Why I Picked This Topic

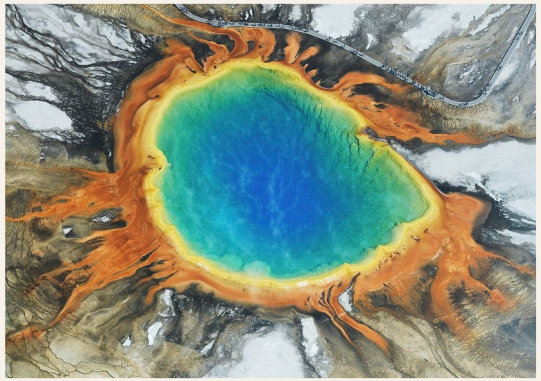
- I have always liked anything related to nature, so when I got this national park dataset I was instantly interested.
- National parks are home to thousands of species, but not all of them are being tracked properly.
- The question I wanted to answer: which category of species is most often listed as protected?
- I hope the result can be a starting point for deciding where to focus monitoring.



*Great Smoky Mountains*

# The Data I Used

- species_info.csv — 5,824 rows. It holds the category, scientific name, common names, and conservation status of each species.
- observations.csv — 23,296 rows. Sightings of each species over a 7-day period.
- The data comes from the National Park Service and is used here as project practice.



Yellowstone



Yosemite



Bryce Canyon



Great Smoky Mts.

# How I Worked On It

1. Opened the data and checked the row counts and column types.
2. Filled in the empty values in the conservation_status column to keep things tidy.
3. Calculated the proportion of protected species for each category.
4. Ran a chi-square test to see whether the difference was real or just chance.
5. Compared the total observations between parks.
6. Pulled everything together into a conclusion.

# Finding 1 — Conservation Status

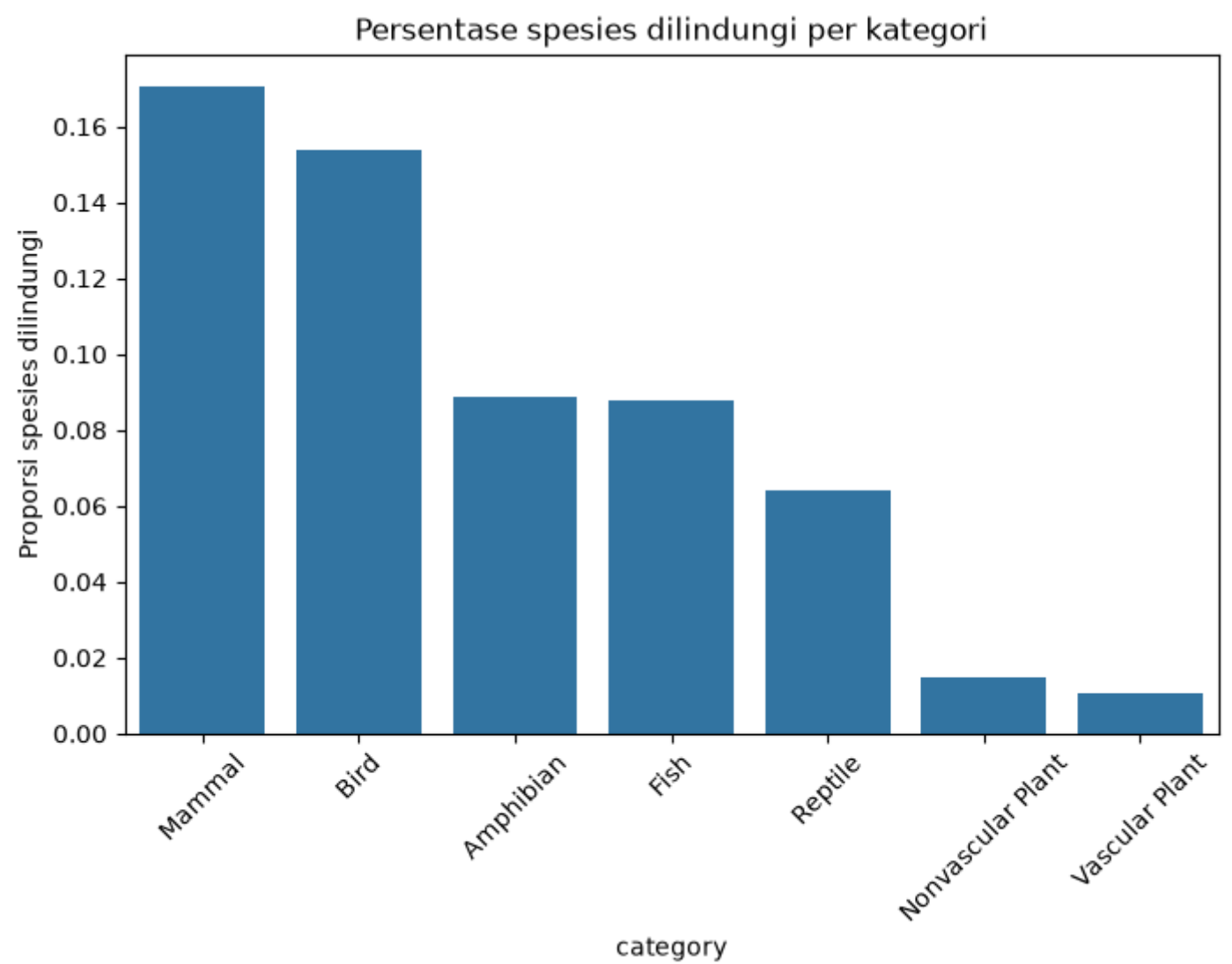
| Status | Count |
|---|---|
| No Intervention | 5,633 |
| Species of Concern | 161 |
| Endangered | 16 |
| Threatened | 10 |
| In Recovery | 4 |

**191**

species are listed as protected, only about 3.3% of the 5,824 species.

The largest group by far is not part of any conservation program at all.

# Finding 2 — Share of Protected Species




Mammals (17.0%) and birds (15.4%) come out on top.

Amphibians and fish sit around 9%.

Plants are at the bottom, only about 1%.

Animals seem to be hit faster by environmental change.

# Finding 3 — Chi-Square Test

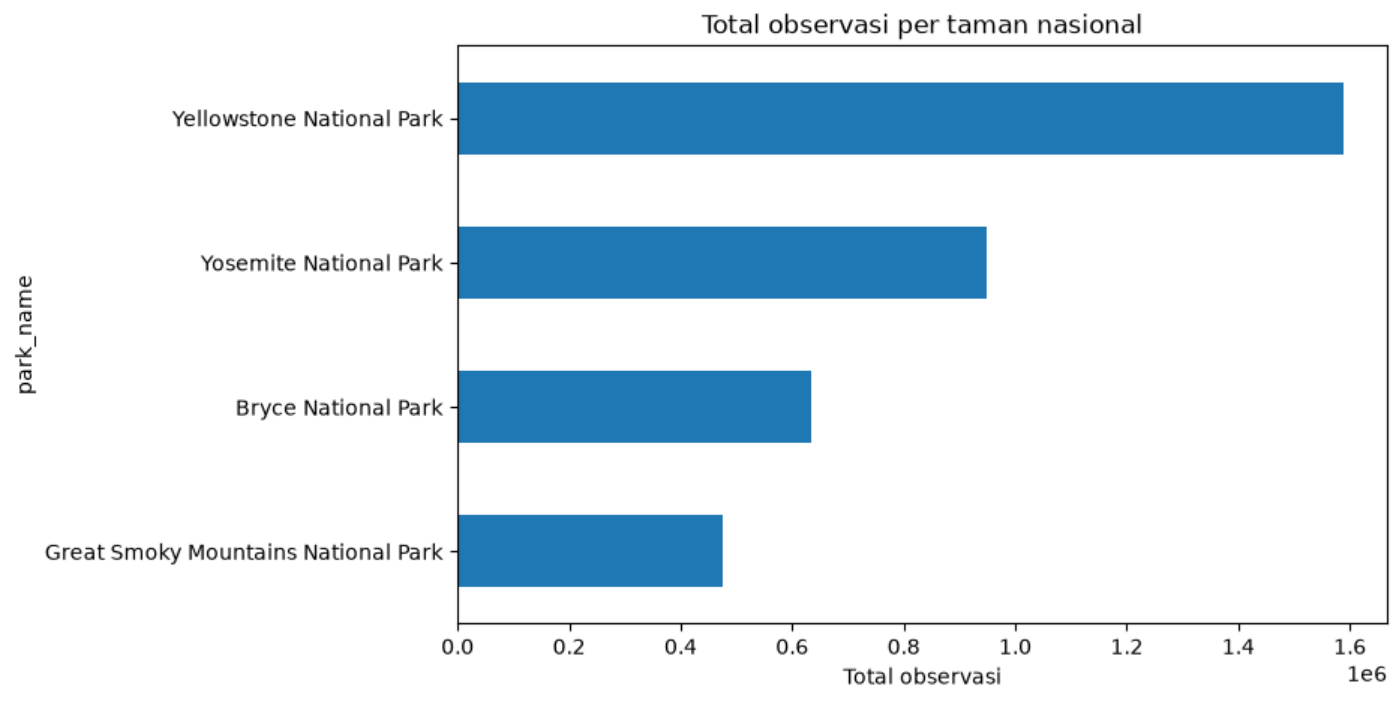
chi2

**628.58**

p-value ≈ $2.46 \times 10^{-117}$

- The p-value is far below 0.05.
- That means species category and conservation status are genuinely related.
- So the pattern I saw earlier is not just a coincidence.

# Finding 4 — Observations per Park




Yellowstone is highest, with 1,587,641 observations.

Followed by Yosemite (949,452) and Bryce Canyon (633,291).

Great Smoky Mountains is the lowest, at 474,863.

This may come down to park size and how much surveying is done.

# Conclusion

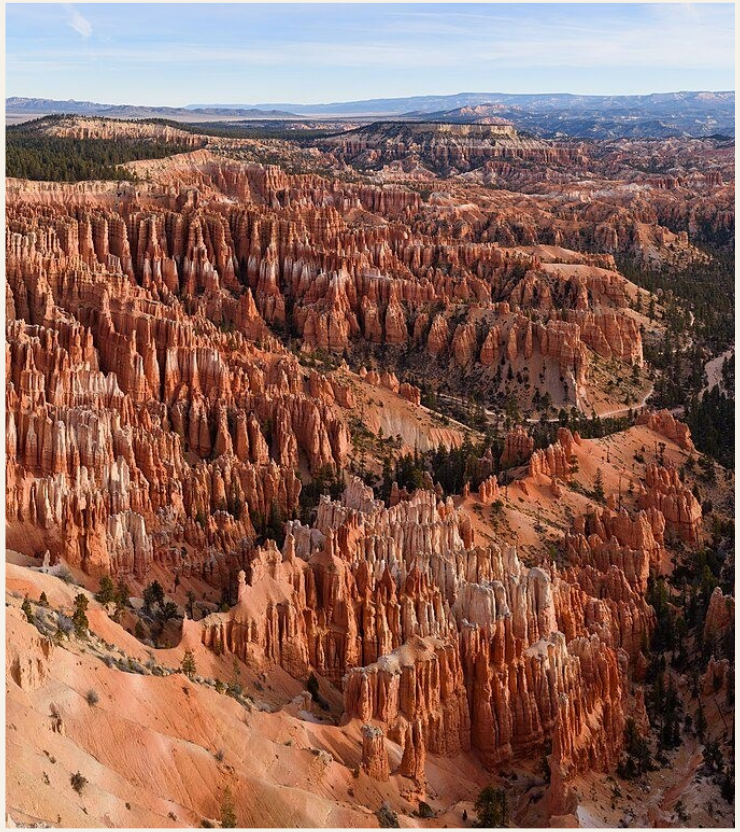
- Most species are not part of any conservation program; among those that are, Species of Concern is the most common status.
- Mammals and birds have the highest share of protected species, plants the lowest.
- The chi-square test shows a significant link between species category and conservation status ($p < 0.05$).
- Yellowstone recorded the highest total number of observations.



*Yosemite National Park*

# Recommendations

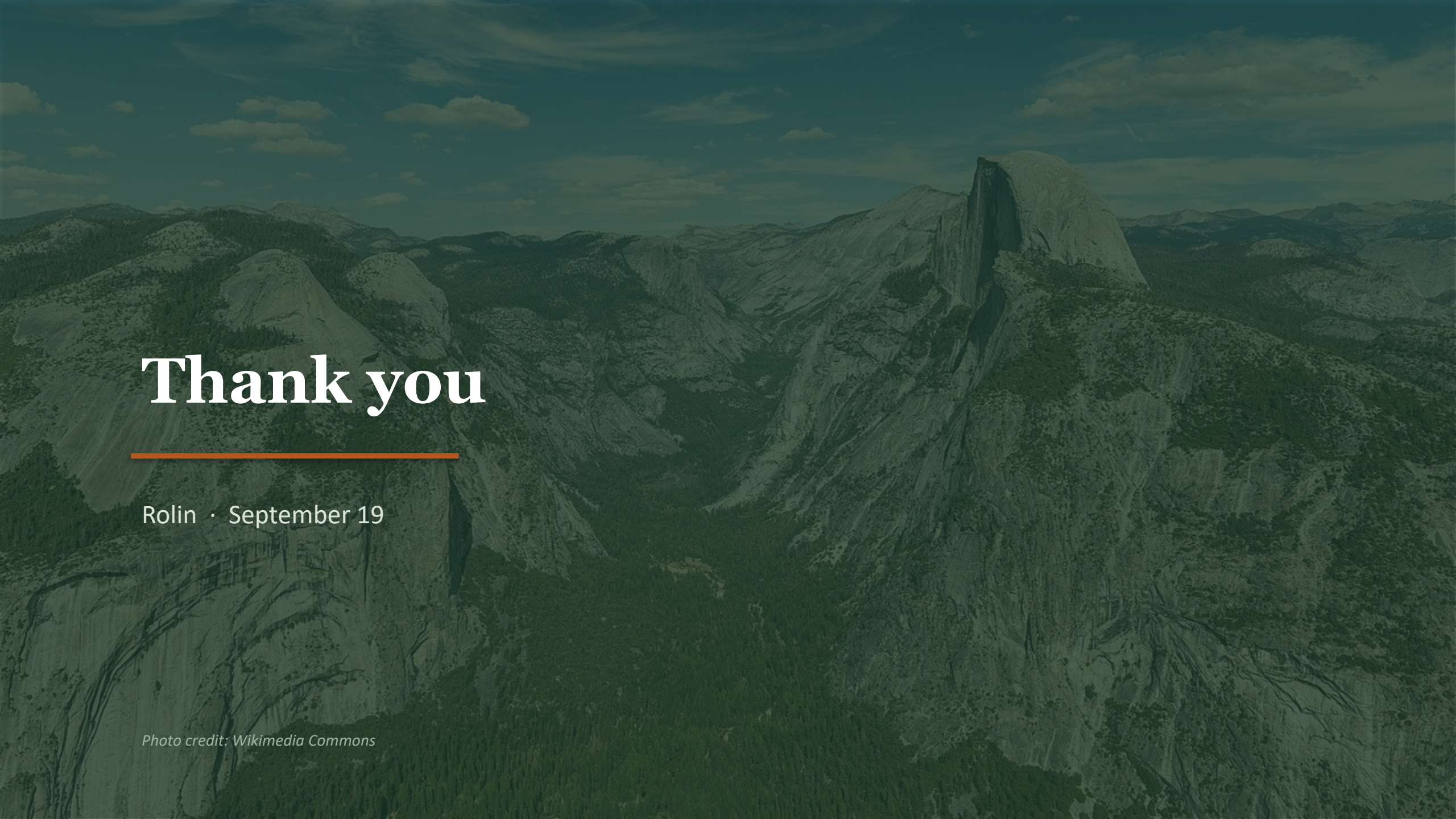
- Conservation efforts could be focused on mammals and birds.
- Monitoring should be increased in the parks with the fewest observations, such as Great Smoky Mountains.
- A period longer than 7 days would likely give a more stable pattern.
- Conservation status data should be refreshed regularly to stay accurate.



*Bryce Canyon National Park*

# Notes & Sources

- Data: National Park Service — Codecademy project dataset.
- Park photos: Wikimedia Commons (public domain / free licence).
- Built with Python (pandas, matplotlib, seaborn) in Jupyter Notebook.
- These slides were put together by me from the analysis in the notebook.



# Thank you

Rolin · September 19

*Photo credit: Wikimedia Commons*